User Churn Project | Waze
Hypothesis Testing
Executive Summary

Overview

As part of Waze app user churn project, our data team was tasked to conduct hypothesis testing. This report shows the procedure we followed and key insights delivered from the statistical analysis.

Problem

We discovered through descriptive statistics that iPhone users has more drives on average compared to android users.

Solution

To investigate whether this relation is statistically significant or due to chance alone, our team started performing two-sample t-test to reveal the relation between these variables.

Details

Test Steps:

1. At first we state the null and alternative hypothesis as follows:

   Null hypothesis: There is no effect of the user's device type on number of rides.

   Alternative hypothesis: There is an effect of the user's device type on number of rides.

2. After setting a significance level of 5% and performing the test we concluded that we reject the null hypothesis.

Test Results:

There is no statistically significant difference in average number of drives between drivers who use iPhone devices and drivers who use Androids. Hence, device type is not an important factor in developing our model.

Next Steps

- Our next step is to perform regression analysis to discover hidden relationships between variables in our data. Neglecting the effect of device type.